

GigEase MLOps Implementation Levels:-

Maturity staged from Level 0 (fully manual) to Level 3 (fully automated continuous training, monitoring, and governance), following the standard MLOps maturity model applied to the GigEase platform.

Level	Characteristic	What It Means	GigEase Example (from the platform docs)
Level 0 Manual Process	Manual, notebook-driven workflow	Data scientists manually pull data, train, and evaluate in notebooks with no orchestration	Income baseline and fraud rules first prototyped by hand in notebooks against historical events like Cyclone Dana
	No automated testing or validation	Data and model quality checked ad hoc, if at all, before use	Multi-source disagreement between NewsAPI, NDMA, Maps, and Police was caught manually, not by an automated gate
	Manual deployment	Engineer manually copies model artifacts to the serving environment	A retrained STFI/RSMD threshold set is manually pushed live with no staged rollout
	Static model, rare retraining	Model is trained once and rarely updated unless something breaks	Fraud rules tuned to one disruption's signature (e.g., one cyclone) go stale until someone notices
Level 1 ML Pipeline Automation	Automated, scheduled training pipelines	Training runs on a fixed cadence via an orchestrated pipeline, not by hand	Income model retrains weekly, risk model fortnightly, and fraud model quarterly through scheduled pipeline jobs
	Automated data & model validation	Pipeline gates reject bad data or underperforming models before promotion	A retrain is blocked automatically if weather data is over 15 minutes stale or GPS pings are missing
	Reusable, modular pipeline components	Shared steps (ingestion, feature build, train, evaluate) reused across models	The same feature-build step produces zone-risk score and peer-income ratio for both the income and fraud pipelines
	Manual or semi-manual deployment	Deployment still requires a human trigger even though training is automated	A validated model version still needs an engineer to approve promotion to production
	CI on every code and pipeline change	Automated linting, unit tests, and schema checks run on every pull request	CI blocks merges that break the 2-of-4 source RSMD confirmation logic before they reach the pipeline

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Level 2 CI/CD Pipeline Automation	Automated build & containerization	Every pipeline/model change automatically builds a versioned container image	Each of the 4 agents (Work History, KYC, Fraud, Decision) is rebuilt and versioned automatically on change
	Automated staged deployment	Canary or shadow rollout happens automatically once tests pass	New risk-model versions automatically shadow-score live traffic before replacing the production model
	Automated champion/challenger promotion	A challenger only replaces the champion if it wins on the defined metric	A challenger fraud model is auto-promoted only if it beats the champion on the cost-weighted objective
Level 3 Full CT, Monitoring & Governance Automation	Drift-triggered continuous training	Retraining fires automatically off live performance signals, not just a calendar	A retrain and rollback trigger automatically when $AUC < 0.75$ or $RMSE\text{-}drift > 15\%$ is detected
	Automated monitoring & alerting	Live dashboards and alerts cover data freshness, drift, and fairness continuously	Cross-agent traceability logging auto-flags a bad payout back to the exact stage (Work History, KYC, Fraud, Decision) that caused it
	Automated rollback & incident response	A degraded model reverts to the last known-good version without manual intervention	A version breaching drift thresholds is rolled back automatically, with a runbook covering the GPS-degradation edge case
	Automated governance, audit & explainability	Compliance artifacts (explanations, audit trails) are generated and stored automatically	SHAP outputs are logged immutably and payout hashes are auto-anchored on Polygon for regulatory-grade audit
	Closed feedback loop into retraining	Outcomes and appeals automatically feed back into the next training cycle	Appeal outcomes from the 30-day IRDAI escalation path automatically enrich the labeled dataset used for the next retrain